UMESH GAIKWAD

INFRASTRUCTURE SOFTWARE ENGINEER | CLOUD PLATFORMS & AI-ASSISTED OPERATIONS

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PROFESSIONAL SUMMARY

Infrastructure and platform engineer with 4+ years of experience building automation and operating cloud-native systems across AWS and GCP. Built an internal Python/FastAPI-based AWS asset discovery tool using LangGraph, LiteLLM and MCP, integrated with AWS Cloud Control API. Experience spans Kubernetes/EKS, infrastructure automation, CI/CD, production troubleshooting, cloud networking and observability. M.Tech in Computer Science from IISc Bengaluru.

TECHNICAL SKILLS

Programming & APIs: Python, Bash, SQL; Java (JUnit test development); FastAPI, REST APIs
AI-assisted infrastructure tooling: LangGraph, LiteLLM, MCP, AWS Cloud Control API, natural-language infrastructure discovery
Cloud & infrastructure: AWS, GCP, EKS, EC2, EBS, ECR, S3, SQS, IAM/STS, ALB/ELB, VPC
Kubernetes & automation: Kubernetes, Docker, Helm, Karpenter, Bottlerocket, Terraform, Pulumi
CI/CD & GitOps: Jenkins, JCasC, Job DSL, Docker Buildx, GitHub Actions, Bitbucket Pipelines, Argo CD
Distributed systems, observability & networking: Kafka, Prometheus, PromQL, VictoriaMetrics/vmalert, New Relic, DNS, TCP/IP, VPC peering, NACLs, security groups, TLS

PROFESSIONAL EXPERIENCE

Platform Engineer-2 | Sep 2024 – Present
Observe.AI | *Bengaluru, India*

- Built an internal natural-language AWS asset discovery service using Python, FastAPI, LangGraph, LiteLLM and MCP; integrated AWS Cloud Control API to retrieve resource information across AWS accounts.
- Managed day-to-day operations of 15+ Kubernetes clusters, ensuring timely version upgrades and continuously tuning cluster configurations, node capacity and workloads for reliability, performance and cost efficiency.
- Optimized EKS Bottlerocket image preloading with EBS snapshot-based workflows, reducing workload readiness from approximately 10 minutes to 120 seconds (∼80%) and saving $1,200 per day (∼$438K annualized).
- Migrated MongoDB connectivity from private endpoints to AWS VPC peering, reducing connectivity costs by 97% and saving $222 per day (∼$81K annualized).
- Worked on Jenkins CI/CD infrastructure on EKS using Helm, JCasC, Job DSL, Docker Buildx, ECR and Kubernetes build agents.
- Contributed to a Terraform change workflow using Jenkins and Jira to publish plan/review output and gate infrastructure apply operations on approval.
- Right-sized Kubernetes workloads and optimized EKS compute costs by migrating suitable workloads to AWS Graviton, validating ARM64 multi-architecture images, and tuning Karpenter NodePools; reduced infrastructure spend by approximately $10K/month (∼$120K annualized).

DevOps Engineer | Aug 2023 – Sep 2024
Amagi Media Labs | *Bengaluru, India*

- Developed Python tooling integrated with Slack for Kubernetes environment operations, including endpoint discovery and guarded on-demand deployments with validation and an audit trail.
- Automated cloud infrastructure using reusable Pulumi YAML stacks and added Slack visibility for AWS Secrets Manager changes.

Big Data Engineer | Jul 2019 – Apr 2020
Exadatum Software Services Pvt Ltd | *Pune, India*

- Set up Docker-based Spark/Livy with Jupyter and wrote Java/JUnit tests for Kafka/GCS data flows.

EDUCATION

M.Tech, Computer Science | Indian Institute of Science (IISc), Bengaluru | 2021–2023
Relevant coursework: Distributed Systems, Network & Distributed Systems Security, Principles of Distributed Software, Scalable Systems for Data Science.

B.E., Computer Science | MGM's College of Engineering, Nanded | 2015–2019

CERTIFICATIONS

- **Certified Kubernetes Administrator (CKA)** | Credential verification
- Google IT Automation with Python | Credential verification